Viridis Research — certification record

Run `runs/2026-10-29_P2` 2026-10-29 project P2, task P2-T001 engine `b6b87d8bd2c0`
GitHub run: `https://github.com/jdhart81/viridis-engine/actions/runs/40889395791`

Re-certifying the Carbon Continuity Threshold: A Lean Proof of the Two-Pool Regenerative Coupling Condition under a Pinned Toolchain
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Labels awarded by the verifier

Claim	Label	Claim text (as submitted)
C1	**PROVEN**	formal [P2.CarbonContinuity]: For all real a, d, r, p with 0 <= a < 1, 0 <= d < 1, r > 0 and p > 0, there exist L > 0 and D > 0 with L <= aL + rD and D <= pL + dD if and only if (1 - a)(1 - d) <= rp.
C2	**ASSUMED**	conjecture: Observation about the proof, not a sealed claim: the equivalence still holds without the hypotheses 0 <= a, 0 <= d and p > 0. The helper lemmas in the proof file use only a < 1, d < 1 and r > 0.

Sealed statement P2.CarbonContinuity

File `lean/ViridisStatements/P2/CarbonContinuity.lean`, sha256 `a5a6e8c63205b865da0df0c302f1926b48d1874ce1310053da6c1ae3104ffded`, sealed by decision D-004. Axioms used by the proof: Classical.choice, Quot.sound, propext. Candidate sha256 `bfced37f622d1ab9921f2fe7bf02ffa7c88dbdeb1e53d3a250df0ff8f0f94f44`. Listing shown with symbol substitution; the exact bytes are in the repository.

```
import Mathlib

/-!
# P2 · re-certification: the Carbon Continuity threshold

The live Viridis OS kernel "Carbon Continuity ? wildfire recovery threshold diagnostic"
(runtime `viridis.carbon-continuity-threshold.v1`, `src/lib/carbonContinuity.ts` in
viridis-conservation-app) decides whether a two-pool carbon portfolio can be kept from
shrinking across repeated disturbance-recovery cycles. Its formal backing is
`Viridis.CarbonContinuity.carbon_continuity_threshold_iff` in the archived canon release
`releases/2026-08-08/Carbon_Continuity_After_Wildfire` (Zenodo 10.5281/zenodo.21855690,
Lean source sha256 899d3bb1929bac8d1ab6b760caffac428de115489e6fc7c1b2752e0769a77fc1).
Its earlier certificate came from the retired Comparator; this module restates the
theorem as a sealed target so the engine can re-prove it under the pinned toolchain.

Model: one cycle maps a living pool `L` and a durable pool `D` to
`L' = a·L + r·D` and `D' = p·L + d·D`, where `a, d ∈ [0, 1)` are the pools' retention
fractions and `r, p > 0` are the transfer rates between them.

`Statement`: with those parameter ranges, a strictly positive portfolio that no cycle
shrinks (componentwise) exists **exactly when** `(1 − a)(1 − d) ≤ r·p`.
-/

namespace ViridisStatements.P2.CarbonContinuity

/-- Living-pool value after one standardized disturbance-recovery cycle. -/
def stepLiving (a r L D : ℝ) : ℝ := a * L + r * D

/-- Durable-pool value after one standardized disturbance-recovery cycle. -/
def stepDurable (d p L D : ℝ) : ℝ := p * L + d * D

/-- A strictly positive portfolio `(L, D)` that one cycle does not shrink in either pool. -/
def Sustainable (a d r p : ℝ) : Prop :=
  ∃ L D : ℝ, 0 < L ∧ 0 < D ∧ L ≤ stepLiving a r L D ∧ D ≤ stepDurable d p L D

/-- **Sealed target `P2.CarbonContinuity`.** For retention fractions `a, d ∈ [0, 1)` and
positive transfer rates `r, p`, a sustainable portfolio exists exactly at or above the
regenerative coupling threshold `(1 − a)(1 − d) ≤ r·p`. -/
def Statement : Prop :=
  ∀ a d r p : ℝ, 0 ≤ a → a < 1 → 0 ≤ d → d < 1 → 0 < r → 0 < p →
    (Sustainable a d r p ↔ (1 - a) * (1 - d) ≤ r * p)

/-! ## Sanity checks

These pin the meaning of the definitions with concrete numbers. -/

/-- The canon's strict witness: `(a, d, r, p) = (3/5, 4/5, 1/2, 1/5)` with `(L, D) = (1/2, 2/5)`
is above the threshold, holds living carbon exactly and grows durable carbon. -/
```

```
example : (1 - (3/5 : ℝ)) * (1 - 4/5) < 1/2 * (1/5) ∧
    stepLiving (3/5) (1/2) (1/2) (2/5) = 1/2 ∧ (2/5 : ℝ) < stepDurable (4/5) (1/5) (1/2) (2/5) := by
  norm_num [stepLiving, stepDurable]

/-- A witness portfolio makes `Sustainable` true at that point. -/
example : Sustainable (3/5) (4/5) (1/2) (1/5) :=
  ⟨1/2, 2/5, by norm_num, by norm_num, by norm_num [stepLiving], by norm_num [stepDurable]⟩

/-- Below the threshold: `(a, d, r, p) = (0, 0, 1/2, 1/2)` gives `(1 − a)(1 − d) = 1 > 1/4 = r·p`. -/
example : ¬ ((1 - (0 : ℝ)) * (1 - 0) ≤ 1/2 * (1/2)) := by norm_num

end ViridisStatements.P2.CarbonContinuity
```

Other checks

- Lean toolchain leanprover/lean4:v4.28.0, Mathlib 8f9d9cff6bd7; kernel replay by `leanchecker`.
- References: 2 entries, 2 DOIs, 2 resolved, 0 unresolved, 0 unchecked.
- Agent paper: compiled.
- Statement proposals: 0.

How to read this record

ASSUMED: not verified (the default). **PROVEN**: a Lean 4 proof of a statement sealed before the attempt, compiled with the pinned toolchain, matched definitionally to the sealed statement, using only the axioms propext, Classical.choice and Quot.sound, and replayed through the kernel. **TESTED/REPLICATED**: reproduced by the verifier (not yet enabled). **ENDORSED/ADOPTED**: granted by people. Novelty is never a label. **TRIVIAL** marks a sealed statement that one of the engine's probe tactics already closes.

This paper was drafted by an AI research agent (Claude, by Anthropic) running the Viridis research engine; formal proof search may use Aristotle (Harmonic). Justin Hart directs the research program. The labels on this page were assigned by an automated verifier, not by the agent, and only they carry certification. Claims in the paper that are not listed here as PROVEN are ASSUMED.

Reproduce everything from `REPRODUCE.md` in the run folder; the run's files are listed with hashes in `MANIFEST.sha256`.

Re-certifying the Carbon Continuity Threshold: A Lean Proof of the Two-Pool Regenerative Coupling Condition under a Pinned Toolchain

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29 October 2026

Abstract

The live Viridis OS kernel *Carbon Continuity* decides whether a two-pool carbon portfolio, made of a living pool and a durable pool, can be kept from shrinking across disturbance-recovery cycles. One cycle maps (L, D) to $(aL + rD,\ pL + dD)$, with retention fractions $a, d \in [0, 1)$ and transfer rates $r, p > 0$. The kernel's formal core states that a strictly positive portfolio that one cycle does not shrink in either pool exists exactly when $(1-a)(1-d) \le rp$. Until now this result rested on a certificate issued by a checker that has since been retired. We restate the theorem as the sealed statement `P2.CarbonContinuity` and give a short Lean 4 proof of it under Lean v4.28.0 with a pinned Mathlib. The proof is ported from the archived canon release [2]. We say exactly what the theorem settles, namely the threshold logic the kernel applies. We also say what it does not settle: any coefficient value, the adequacy of the linear model, and any ecological or carbon-credit outcome.

1 Introduction

Viridis OS runs a set of decision kernels. Each one is a small piece of software that turns ecological parameters into a decision. The Carbon Continuity kernel (runtime `viridis.carbon-continuity-threshold.v1`) answers one question: after a wildfire or a similar disturbance, can a carbon portfolio be kept from shrinking cycle after cycle? It answers by comparing a *leakage product* $(1 - a)(1 - d)$ with a *regenerative loop gain* rp.

The rule is only as trustworthy as the mathematics behind it. The canon release [2] proved the rule in Lean, but the certificate came from tooling that can no longer be re-run. This note re-proves the rule against a sealed restatement, so the kernel's mathematical assurance can be re-checked with the current toolchain. The question is narrow on purpose. We ask whether the threshold rule is a correct consequence of the stated model. We do not ask whether the model or its coefficients describe any real forest.

2 Background and related work

The cycle map is the linear map $x \mapsto Mx$ with the nonnegative matrix

$$M = \begin{pmatrix} a & r \\ p & d \end{pmatrix}, \qquad x = \begin{pmatrix} L \\ D \end{pmatrix}.$$

Since $r, p > 0$, the matrix M is irreducible. By the Collatz–Wielandt characterisation in Perron–Frobenius theory [1], the spectral radius satisfies $\rho(M) \geq 1$ exactly when some $x > 0$ has $Mx \geq x$. In addition, $\rho(M) \geq 1$ holds, given $a, d < 1$, exactly when $\det(I - M) = (1-a)(1-d) - rp \leq 0$. The theorem proved here is therefore a two-dimensional instance of classical theory. We claim no novelty for the mathematics. The contribution is a re-checkable machine proof tied to a live decision kernel. The canon release [2] gives the ecological framing and the original Lean development, including corollaries that the sealed statement does not include (boundary stationarity, strict growth above threshold).

3 Model and statement

The sealed module defines

$$\text{stepLiving}(a, r, L, D) = aL + rD, \qquad \text{stepDurable}(d, p, L, D) = pL + dD,$$
$$\text{Sustainable}(a, d, r, p) : \iff \exists L, D,\ 0 < L,\ 0 < D,\ L \leq aL + rD,\ D \leq pL + dD.$$

Theorem 1 (`P2.CarbonContinuity`)**.** *For all real a, d, r, p with $0 \leq a < 1$, $0 \leq d < 1$, $r > 0$ and $p > 0$,*

$$\text{Sustainable}(a, d, r, p) \iff (1 - a)(1 - d) \leq rp.$$

4 Proof

Lemma 2 (Sufficiency)**.** *If $a < 1$, $r > 0$ and $(1 - a)(1 - d) \leq rp$, then $(L, D) = (r, 1 - a)$ is a witness.*

Proof. Both components are positive. For the living pool, $ar + r(1-a) = r$, so the inequality holds with equality. For the durable pool, $pr + d(1 - a) - (1 - a) = rp - (1 - a)(1 - d) \geq 0$. □

Lemma 3 (Necessity)**.** *If $a < 1$, $d < 1$ and (L, D) is a witness, then $(1 - a)(1 - d) \leq rp$.*

Proof. The two inequalities rearrange to $(1 - a)L \leq rD$ and $(1 - d)D \leq pL$. Both left-hand sides are positive, so we can multiply the inequalities: $(1 - a)(1 - d)\, LD \leq rp\, LD$. Cancelling $LD > 0$ gives the threshold. □

Lean development. The proof file `proofs/P2.CarbonContinuity.lean` follows the canon proof almost line for line. It is restated against the sealed definitions, and the heavier `nlinarith` calls are replaced by `linarith` where the goal is linear after normalisation.

```
theorem sufficient (a d r p : ℝ) (ha1 : a < 1) (hr : 0 < r)
    (h : (1 - a) * (1 - d) ≤r * p) : Sustainable a d r p := by
  refine ⟨r, 1 - a, hr, by linarith, ?_, ?_⟩
  ·simp only [stepLiving]; nlinarith
  ·simp only [stepDurable]; nlinarith

theorem necessary (a d r p : ℝ) (ha1 : a < 1) (hd1 : d < 1)
    (hs : Sustainable a d r p) : (1 - a) * (1 - d) ≤r * p := by
  obtain ⟨L, D, hL, hD, hliv, hdur⟩ := hs
  simp only [stepLiving] at hliv
  simp only [stepDurable] at hdur
  have h1 : (1 - a) * L ≤r * D := by linarith
  have h2 : (1 - d) * D ≤p * L := by linarith
```

```
  have h3 : ((1 - a) * L) * ((1 - d) * D) ≤(r * D) * (p * L) :=
    mul_le_mul h1 h2 (by nlinarith) (by nlinarith)
  have h4 : ((1 - a) * (1 - d)) * (L * D) ≤(r * p) * (L * D) := by linarith
  exact le_of_mul_le_mul_right h4 (mul_pos hL hD)

theorem main : ViridisStatements.P2.CarbonContinuity.Statement := by
  intro a d r p _ ha1 _ hd1 hr _
  exact ⟨necessary a d r p ha1 hd1, sufficient a d r p ha1 hr⟩
```

5 Results

We give a Lean proof of the sealed statement `P2.CarbonContinuity`. The local pre-check tool reports that the file compiles under the pinned toolchain, matches the sealed statement, and depends only on `propext`, `Classical.choice` and `Quot.sound`. The labels are awarded by the independent verifier, whose certification page comes before this paper.

Remark 4. We observe that the helper lemmas use only $a < 1$, $d < 1$ and $r > 0$. The hypotheses $0 \leq a$, $0 \leq d$ and $p > 0$ are part of the kernel's parameter domain, but the proof does not need them.

6 What the theorem settles and what it does not

- **Settled (subject to verification):** inside the stated linear two-pool model, the kernel's rule "sustainable iff $(1 - a)(1 - d) \leq rp$" is a correct decision procedure, including at the boundary.
- **Not settled:** the values of a, d, r, p for any site. These come from field evidence and stay ASSUMED until a TESTED lane or a human review covers them.
- **Not settled:** whether a linear, time-invariant two-pool cycle model describes post-fire carbon dynamics. The theorem also says nothing about uncertainty in the coefficients, about stochastic disturbance, or about any carbon-credit or treatment outcome.
- **Not settled:** the statement is about one cycle. Persistence over many cycles follows informally because M is monotone on nonnegative vectors, so $Mx \geq x$ implies $M^k x \geq x$. That fact is not part of the sealed statement.
- **Not settled:** that the TypeScript implementation (`src/lib/carbonContinuity.ts`) implements this rule faithfully. Code conformance has not been checked here.

7 Limitations

The result is mathematically elementary. Its value lies in being re-checkable and tied to a deployed kernel, not in its depth. The canon corollaries (boundary stationarity, strict growth, nonvacuity) were not re-proved as sealed statements. The sealed module includes concrete sanity examples that pin the meaning of the definitions.

8 Reproducibility

Toolchain `leanprover/lean4:v4.28.0`, Mathlib commit `8f9d9cff6bd728b17a24e163c9402775d9e6a365`. Sealed statement source sha256 `a5a6e8c63205b865da0df0c302f1926b48d1874ce1310053da6c1ae3104ffded`. Canon source sha256 `899d3bb1929bac8d1ab6b760caffac428de115489e6fc7c1b2752e0769a77fc1` (canon repository commit `80d32904d751b45f415e23f872510cc2eddca3ff`). To check the proof, run `viridis-lean-check proofs/P2.CarbonContinuity.lean P2.CarbonContinuity`. No automated prover jobs and no external data were used.

9 Changelog

- 2026-10-29 (P2-T001): First version. Ported the canon threshold proof to the sealed statement `P2.CarbonContinuity`, and wrote the scope statement.

References

[1] Abraham Berman and Robert J. Plemmons. *Nonnegative Matrices in the Mathematical Sciences*, volume 9 of *Classics in Applied Mathematics*. SIAM, Philadelphia, PA, 1994. doi: 10.1137/1.9781611971262.

[2] Justin D. Hart. Carbon continuity after wildfire: A dynamic portfolio framework and regenerative coupling threshold for natural carbon assets. Zenodo, version 0.2.0-formal, August 2026.